

make the contents for all the topics listed in java

ChatGPT said:

JAVA – Collections + Applets + AWT **Complete Notes**

1. Collections Framework

1.1 Overview

The **Java Collections Framework (JCF)** is a set of classes and interfaces that store and manipulate groups of data (objects).

Advantages

- Ready-made data structures
- Reduces coding effort
- High performance
- Algorithms (sorting, searching) built-in
- Unified architecture (same interface, different implementations)

1.2 Key Interfaces

Interface	Description
Collection	Root interface for all collections
List	Ordered collection, duplicates allowed
Set	No duplicates
Map	Stores key–value pairs
Queue	Follows FIFO

● 2. Important Collection Classes

2.1 ArrayList

- Dynamic array
- Fast for **access**, slow for **insertion/deletion** in the middle
- Maintains insertion order

Example

```
import java.util.*;

class Demo {
    public static void main(String[] args) {
        ArrayList<String> list = new ArrayList<>();
        list.add("A");
        list.add("B");
        list.add("C");
        System.out.println(list);
    }
}
```

2.2 LinkedList

- Doubly linked list
- Fast insertion/deletion
- Good as **List + Queue**

```
LinkedList<Integer> ll = new LinkedList<>();
ll.add(10);
ll.addFirst(5);
ll.addLast(20);
System.out.println(ll);
```

2.3 HashSet

- No duplicates
- No order preserved
- Backed by hashing (HashMap)

```
HashSet<Integer> hs = new HashSet<>();
hs.add(10);
hs.add(20);
```

```
hs.add(10); // ignored
System.out.println(hs);
```

2.4 Map (HashMap Example)

- Key-Value pairs
- Keys must be unique

```
HashMap<Integer, String> map = new HashMap<>();
map.put(1, "A");
map.put(2, "B");
System.out.println(map.get(1));
```

3. Applets

 **Applet is deprecated in Java 9+**, but still taught in syllabus.

3.1 What are Applets?

Applets are **Java programs** that run **inside a web browser** or **applet viewer**, not inside JVM normally like console apps.

3.2 The Applet Class

Located in:
`java.applet.Applet`

Important Methods:

- `init()`
 - `start()`
 - `stop()`
 - `destroy()`
 - `paint(Graphics g)`
-

3.3 The Applet and HTML

To load an applet using HTML:

```
<applet code="MyApplet.class" width="300" height="300">
</applet>
```

4. Life Cycle of an Applet

Phase	Method	Description
Initialization	<code>init()</code>	Called once when applet loads
Start	<code>start()</code>	Begins execution
Running	<code>paint(Graphics g)</code>	Draws components
Stop	<code>stop()</code>	Applet paused
Destroy	<code>destroy()</code>	Cleanup before closing

Applet Example

```
import java.applet.*;
import java.awt.*;

public class MyApplet extends Applet {
    public void init() {
        setBackground(Color.yellow);
    }

    public void paint(Graphics g) {
        g.drawString("Hello Applet!", 50, 50);
    }
}
```

5. The Graphics Class

Used for drawing shapes/text.

Basic Methods

```
g.drawString("Hello", x, y);
g.drawLine(x1, y1, x2, y2);
g.drawRect(x, y, w, h);
g.fillOval(x, y, w, h);
g.setColor(Color.red);
```

6. Painting the Applet

Painting is always done inside:

```
public void paint(Graphics g)
```

Example:

```
public void paint(Graphics g) {  
    g.setColor(Color.red);  
    g.drawOval(50, 50, 100, 100);  
}
```

7. User Interfaces for Applet

Applets use **AWT Components** such as:

- Button
 - Label
 - TextField
 - Checkbox
 - Choice (dropdown)
 - TextArea
-

Example: Adding Components

```
public void init() {  
    Button b = new Button("Click Me");  
    add(b);  
  
    TextField tf = new TextField(20);  
    add(tf);  
}
```

8. AWT Controls

Component	Usage
Button	Clickable button

Component	Usage
Label	Non-editable text
TextField	Input field
TextArea	Multi-line text
Checkbox	True/False
Choice	Drop-down list
List	Scrollable list
Scrollbar	Slider control

Example Using Multiple AWT Controls

```
import java.applet.*;
import java.awt.*;

public class UIApplet extends Applet {
    public void init() {
        Label l = new Label("Name : ");
        TextField tf = new TextField(20);
        Button b = new Button("Submit");

        add(l);
        add(tf);
        add(b);
    }
}
```

Everything Covered

- ✓ Collections overview
 - ✓ Interfaces
 - ✓ ArrayList, LinkedList, HashSet, Map
 - ✓ What are applets
 - ✓ Applet class
 - ✓ Applet and HTML
 - ✓ Applet life cycle
 - ✓ Graphics class
 - ✓ Painting applet
 - ✓ UI for applet
 - ✓ AWT controls
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